Local/Global Optimality

$M = (E, \mathcal{I})$, w a weight function.

<u>Thm</u>: $B \in \mathcal{I}$ is a base, it is max weight

iff $w(B') \le w(B)$ for any

base B', with $|B' \setminus B| = 1$.

pf: Suppose $\exists B'$ with $w(B') > w(B)$.

$\exists$ perfect matching N on $B \triangle B'$ on $A(B)$.

Since $w(B') > w(B)$, $\exists$ edge (y, z) in N

with $w(z) > w(y)$.

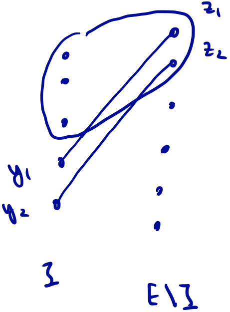
$\Rightarrow$ $B - y + z$ is a base with $w(B - y + z) > w(B)$

Using the Bipartite Graph

Thm: Let $M = (E, \mathcal{I})$, let I, J s.t. $I \in \mathcal{I}$, and $|I| = |J|$.

If $A(I)$ contains a unique P.M. N on $I \triangle J$, then $J \in \mathcal{I}$.

Picture:



Recall: edge (y,z) means

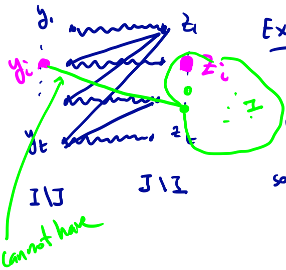
$$I - y + z \in \mathcal{I}.$$

$$I - y_1 + z_1 \in \mathcal{I}$$

$$I - y_2 + z_2 \in \mathcal{I}$$

Want: $\boxed{I - y_1 - y_2 + z_1 + z_2 \in \mathcal{I}}$.

Proof of Theorem



Exercise: Since N is the unique matching in $A(I)$ we can order the pairs (y_i, z_i) so that y_i has edges to z_j only for $j \le i$.

If $J \notin \mathcal{I} \Rightarrow \exists\, C \subseteq J$ a circuit. $C \cap (J \setminus I) \neq \emptyset$

$\Rightarrow C \cap \{z_1, \dots, z_t\} \neq \emptyset$. Choose smallest i s.t. $z_i \in C$.

$\Rightarrow (y_i, z) \notin A(I) \quad \forall z \in C \setminus I - z_i$

$\Rightarrow z \in \operatorname{Span}(I - y_i) \quad \forall z \in C \setminus I - z_i$

$\Rightarrow (C \setminus I - z_i) \subseteq \operatorname{Span}(I - y_i) \Rightarrow \underline{(C - z_i)} \subseteq \operatorname{Span}(I - y_i)$

$z_i \in \operatorname{Span}(C - z_i) \Rightarrow \boxed{z_i \in \operatorname{Span}(I - y_i)}$

This is a contradiction. $(y_i, z_i) \in A(I) \iff I - y_i + z_i \in \mathcal{I}$. ∎

Another Corollary

Corollary: $M = (E, \mathcal{I})$ a matroid.

$I \in \mathcal{I}$, $J \subseteq E$, $|I| = |J|$, and $r(I \cup J) = r(I)$

Suppose $A(I)$ contains a unique P.M. N

on $I \triangle J$.

Let $s \notin I \cup J$ s.t. $I + s \in \mathcal{I}$.

Then: $J + s \in \mathcal{I}$.

Note: The prev. corollary tells us $J \in \mathcal{I}$.

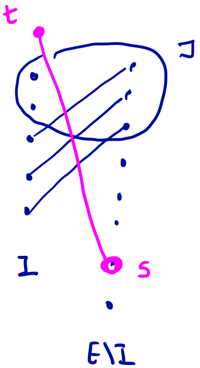
Exercise: Without the assumption $r(I \cup J) = r(I)$

this result could fail.

Proof of Corollary

We create a new element t

$M' = (E \cup \{t\}, \mathcal{I}')$, $F \in \mathcal{I}'$ iff $F \setminus t \in \mathcal{I}$



Consider: $D_{M'}(I \cup \{t\})$

Since: $I + j \notin \mathcal{I}$ $\forall j \in J \setminus I$

there is no arc from t to $J \setminus I$

However, there is an arc from t to s, since $I + s \in \mathcal{I}$.

Hence $N \cup \{(t,s)\}$ is a unique P.M.

on $I \triangle J \cup \{s, t\}$

Hence, by prev. corollary, $J \cup s \in \mathcal{I}'$

$\Rightarrow J \cup s \in \mathcal{I}$. $\boxtimes$.